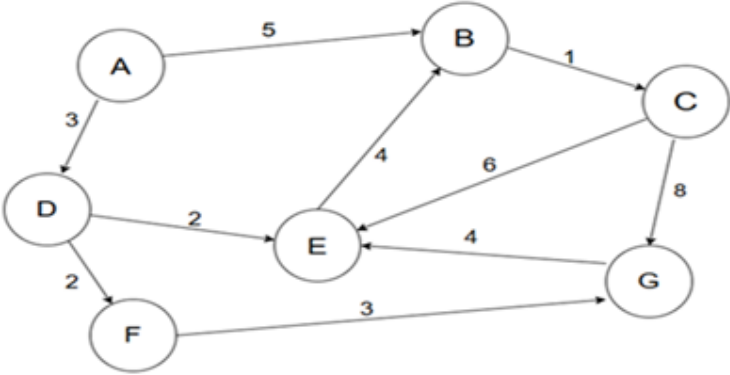
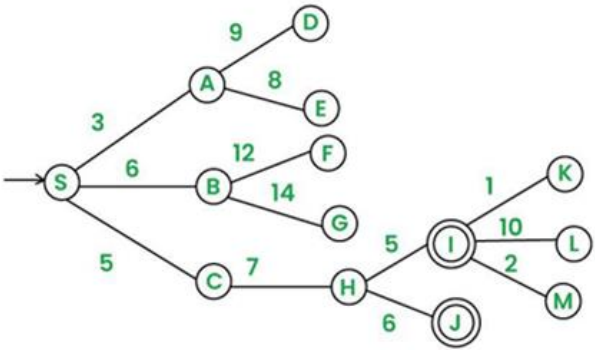
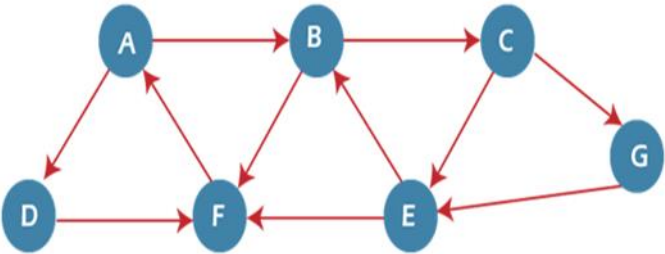


Sr. No.	Questions
1	What is Artificial Intelligence (AI)?
2	What are the primary goals of AI? Define narrow AI and general AI.
3	Mention a few historical milestones in the development of AI.
4	Define intelligence and its types.
5	Explain the concept of agents and environments in AI.
6	Provide examples of AI applications in various fields.
7	Name some sub-areas of AI research.
8	What are some challenges in implementing AI technology in real-world scenarios?
9	How does AI impact society?
10	Describe the categorization of AI systems based on their functionality, providing examples of each.
11	Explore the concept of agents in AI, detailing their characteristics and roles within environments.
12	Provide an overview of significant AI applications in sectors such as healthcare, finance, transportation, and entertainment.
13	Delve into the sub-areas of AI research, elucidating their objectives and methodologies.
14	Analyze current trends in AI, considering advancements in machine learning, natural language processing, robotics, and ethical AI development.
15	Assess the ethical implications of AI adoption in society, considering issues like bias, privacy, and job displacement.
16	Discuss PEAS DESCRIPTORS for i) Part picking robot ii) Self Driving Car iii) Autoamted english tutor iv)Refinery Controller v)Medical Diagnosis System
17	What are some challenges in implementing AI technology in real-world scenarios?
18	Explore sub-areas within AI research, such as computer vision, robotics, and autonomous systems, discussing their significance and potential applications.
19	Discuss future prospects and challenges in AI research and development, considering emerging technologies and societal implications.
20	Analyze current trends in AI, including advancements in deep learning, reinforcement learning, and transfer learning.
21	Compare and contrast different types of AI intelligence, including weak AI, strong AI, and superintelligent AI.
22	Explain what is EDA and why is it important in data analysis?
23	Discuss how can missing values in a dataset be handled effectively during data preprocessing?
24	Explain what is conditional filtering in Pandas and how is it applied?
25	Explain how can you create a Pandas DataFrame?
26	Discuss some common file I/O operations in Pandas?
27	Explain how do you merge Pandas DataFrames?
28	What are join operations in Pandas and how are they performed?
29	What are the key steps involved in EDA using Python programming?
30	Explain the type of EDA with example.

31	Explain how do you typically understand and examine a dataset before performing analysis?
32	Explain the concept of groupby in Pandas and its significance in data analysis.
33	Describe some common challenges encountered during EDA and how to address them?
34	Explain join operations in Pandas and how are they performed?
35	Discuss some common methods for dealing with categorical variables during EDA?
36	Explain implications of imputing missing values versus removing them from a dataset?
37	What are some potential challenges one might encounter during EDA and how can they be addressed?
38	Outline the steps involved in performing EDA using Python programming?
39	Describe the process of understanding and looking at the data before performing EDA. What are some common methods used to gain insights into the structure and characteristics of a dataset?
40	Explain the role of Pandas Series in EDA. How do you access data in a series, and what are some common operations performed on series data?
41	Discuss the file I/O operations commonly used in Pandas for reading and writing data to different file formats. Provide examples of how to read and write data using Pandas.
42	Explain the Pandas operations of indexing, conditional filtering, applying functions, groupby, sorting pivot tables, merging DataFrames, concatenation, and join operations
43	What are some common issues encountered in designing search programs?
44	Define BFS and explain its key characteristics.
45	What does DFS stand for, and how does it differ from BFS?
46	Explain the concept of heuristic search techniques.
47	What is Generate-and-Test, and how is it used in problem-solving?
48	Describe the Hill Climbing algorithm briefly.
49	What makes Best-First Search a popular heuristic search algorithm?
50	Consider the following graph represented by an adjacency list: Graph: A -> B, C B -> D, E C -> F D -> G E -> H F -> I Perform Depth-First Search (DFS) starting from node A and goal node is I.

51	Find the shortest path from node A to node G using Uniform Cost Search. 
52	Solve Using Best First Search. 
53	Perform Breadth First Search.  Adjacency Lists - A : B, D - B : C, F - C : E, G - D : F - E : B, F - F : A, E - G : E
54	Formulate Problem for a Water Jug Scenerio.
55	Explain the A* algorithm, highlighting its key components and how it combines the advantages of both BFS and heuristic search.
56	What is the Alpha-beta search algorithm, and how does it improve upon the minimax algorithm in game playing?
57	Describe the problem reduction approach in problem-solving and provide an example scenario where it can be employed.
58	Describe AO* algorithm, and how does it differ from the A* algorithm?
59	Explain the concept of Local Beam Search and discuss its advantages in solving certain types of problems.
60	Explain Minimax Search algorithm, and how is it used in game theory?
61	Discuss the Hill Climbing algorithm, including its basic mechanism and its suitability for solving optimization problems.
62	How does the Best-First Search algorithm work, and what makes it suitable for searching in large state spaces?

63	Compare and contrast Breadth-First Search (BFS) and Depth-First Search (DFS) algorithms in terms of their search strategies and applications.
64	What does CSP stand for in the context of problem-solving?
65	Define backtracking search briefly?
66	What is the N-Queens problem concerned with?
67	Explain the concept of map coloring.
68	What is the main objective of backtracking search?
69	Describe the process of variable ordering in backtracking search.
70	What is Crypt Arithmetic?
71	Discuss the concept of map coloring in detail, including its significance and challenges it poses in constraint satisfaction problems.
72	Solve using Crypt Arithmetic: $MATH + MYTH = HARD$
73	Explain the backtracking search algorithm step by step, illustrating its working with a concrete example.
74	Describe the N-Queens problem, including its background, objectives, and possible solution strategies.
75	What are the main challenges faced in solving CSPs?
76	How is the solution space represented in the N-Queens problem?
77	Explain the representation of the atomic state in problem-solving?
78	Explain the steps involved in solving the N-Queens problem using backtracking search, highlighting the key decisions made at each stage.
79	Compare and contrast backtracking search with other search algorithms commonly used in problem-solving tasks.
80	Solve using Crypt Arithmetic: $XENIUM + ATOMIC = ELEMENT$
81	Elaborate on the techniques used to represent the solution space in the N-Queens problem and their implications for the search process.
82	How does backtracking search deal with infeasible or partial solutions during its execution?
83	Discuss the significance of variable and value ordering strategies in improving the efficiency of backtracking search, citing relevant examples.
84	Explain Crypt Arithmetic, and how can it be formulated as a constraint satisfaction problem? Provide an example to elucidate.
85	What is a knowledge-based agent?
86	Represent the following statements using FOL. 1. Everyone who sees Mary loves Mary 2. Every boy who loves Mary hates every other boy who Mary loves.
87	What is propositional logic and write syntax of first-order logic (FOL)?
88	What is unification?
89	What techniques are used for representing knowledge in uncertain domains?
90	How do we represent uncertain knowledge show with example?
91	What are the semantics of a belief network?
92	Explain the premise of the Wumpus World environment?
93	Solve using cryptarithmic: $SOME + TIME = SPENT$
94	Can you explain the concept of the Wumpus World?
95	Explain process of forward chaining?
96	Explain backward chaining briefly.
97	How is knowledge engineered in first-order logic?
98	Discuss the role and functioning of knowledge-based agents in artificial intelligence, providing examples to illustrate their applications.

99	Describe the components and rules of the Wumpus World environment, outlining how an agent navigates and solves problems within it.
100	Explain the concepts of syntax and semantics in propositional logic, illustrating with examples to demonstrate their significance.
101	Elaborate on the syntax and semantic aspects of first-order logic, highlighting its expressive power and limitations compared to propositional logic.
102	Explain Unification algorithm. For each pair of the following atomic sentences find the most general unifier (If exist) Find the MGU of $\{p(f(a), g(Y)) \text{ and } p(X, X)\}$ Find the MGU of $\{p(b, X, f(g(Z))) \text{ and } p(Z, f(Y), f(Y))\}$ Find the MGU of $\{p(X, X), \text{ and } p(Z, f(Z))\}$ Find the MGU of UNIFY(prime(11), prime(y)) UNIFY(knows(Richard, x), knows(Richard, John))
103	How is knowledge engineering carried out in the context of first-order logic? Discuss the process and challenges involved, providing examples where applicable.
104	Explain the concept of unification in logic, detailing its role in predicate logic and its application in theorem proving.
105	Explore the concept of uncertainty in knowledge representation, highlighting the challenges it poses and the techniques used to address it.
106	What constitutes the planning problem in AI?
107	How is planning with state space search typically approached?
108	What is partial order planning?
109	What are rule-based systems in expert systems?
110	What does conditional planning entail?
111	What are the different forms of learning in AI?
112	Define inductive learning.
113	What is a decision tree in the context of learning?
114	Describe each component in the architecture of Expert System? What are the limitations of an Expert System?
115	Outline the phases involved in building an expert system?
116	Describe the architecture of an expert system.
117	Explain the role of the inference engine in an expert system.
118	Describe Match-Resolve-Execute in expert systems?
119	Explain the concept of hierarchical planning with example
120	Discuss the various approaches to solving the planning problem, including planning with state space search, partial order planning, hierarchical planning, and conditional planning, highlighting their strengths and weaknesses.
121	Explain how state space search is used in planning, outlining the search algorithms commonly employed and discussing their applicability to different types of planning problems.
122	Describe the concept of partial order planning, including its representation of plans and the methods used to generate and execute plans.
123	Discuss hierarchical planning in detail, explaining how it decomposes complex tasks into simpler subtasks and how it facilitates efficient planning.
124	Explain the process of conditional planning, including the representation of preconditions and effects, and the methods used to generate plans based on varying conditions.
125	Compare and contrast different forms of learning in AI, including supervised learning, unsupervised learning, and reinforcement learning, providing examples of each.

126	Discuss Reinforcement Learning. Give its type.
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